Mu A Individual

1. What is $f'(0)$ if $f(x) = \begin{cases} e^x, & x \leq 0 \\ e^{-x}, & x > 0 \end{cases}$?

2. Mariza is a professional stunt driver, and is filming for a scene where she drives on a straight track. She drives in reverse for the first five minutes of filming. Her position is represented by the function: $S(t) = -0.005t^3$ where t is minutes after filming begins, and S is her position in meters. What is Mariza's speed after 2 minutes of filming?

3. After driving in reverse for 5 minutes, Mariza decides to accelerate forward at 0.03 meters/min^2 for 5 minutes. Is the function for her position differentiable on the interval between 1 minute and 10 minutes of filming?

4. Find $d^6/dx^6\, f(x)\ \ if\ f(x) = xe^x$.

5. The absolute value of $f'(x)$ is $12x^3$. What are two of the possible functions of $f(x)$ if the y −intercept of $f(x)$ is 2?

6. Find

$$\lim_{h\to 0} \frac{\cos\left(\frac{\pi}{3} + h\right) - \frac{1}{2}}{h}$$

7. Find the derivative of the following polynomial: $e^5 + 7$.

8. Find the limit:

$$\lim_{x\to 1} \frac{x}{ln(x)}$$

9. Find the equation of the line tangent to $y = -3x^2 + 2$ at $(2, -10)$

10. Find

$$\frac{d}{dx} sin^{-1}(3x)$$

11. Find $\lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$ for $f(x) = \pi^x, x > 0$

12. For what values of x on the interval $[0,2\pi]$ is the line tangent to $f(x) = x - 2cos(x)$ horizontal?

13. Find the second derivative of $f(x) = 2e^{-6x}$

14. Calculate $\frac{d}{dx}\sqrt{5x^2 + 1}$ when $x = 1$

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15. Find the tangent line at (1,1) of $x^2 + xy + y^2 = 3$.

16. Find $\lim_{x \to \pi} \frac{\cos(x)+\sin(x)}{\cos(-x)}$.

17. Find $\frac{d^2y}{dx^2}$ $of\ y = 4\sin(3x)$

18. $\lim_{x \to \infty} \frac{1-x}{\cos(x)}$

Benjamin Franklin Solutions

1. DNE
2. 0.060 m/min
3. "Yes"? I think.
4. $xe^x + 6e^x$
5. $y = 3x^4 + 2$ and $u = -3x^4 + 2$
6. $\frac{-\sqrt{3}}{2}$
7. 0
8. DNE
9. $y = -12x + 14$
10. $\frac{3}{\sqrt{1-9x^2}}$
11. $\ln(\pi) * \pi^x$
12. $\frac{7\pi}{6}$ and $\frac{11\pi}{6}$
13. $f''(x) = 72e^{-6x}$
14. $\frac{5\sqrt{6}}{6}$
15. $y = -x + 2$
16. 1
17. $y'' = -36sin(3x)$
18. DNE